Taking the beat of the UNSeEN: the Ultraluminous NS populatioN of ULXs

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1. Abstract

Owing to their X-ray luminosity far in excess of the Eddington limit ($L_{\rm Edd}$) for a stellar mass black hole, (BH, $\sim$10 $M_\odot$), ultraluminous X-ray sources (ULXs) have long been considered a very well suited population to look for and study BHs of intermediate mass (IMBHs). The recent discovery of three ULXs showing fast ($\sim$ 1 s) and rapidly evolving pulsations (PULXs) unambiguously associated them to neutron stars (NSs) exceeding by more than 500 times their $L_{\rm Edd}$. These discoveries challenge our understanding of accretion physics and pose a key question about the nature of the ULXs as a class: are ULXs a heterogeneous population hosting both accreting BHs and NSs, or is there a dominant new *Ultraluminous Neutron Star Extragalactic populatioN* (UNSeEN) with extreme properties? **The present proposal involves a large fraction of the ULX community and is aimed at answering the question by tripling the number of ULXs over which sensitive searches for X-ray pulsations and cyclotron resonant scattering features (CRSFs) can be carried out. We expect to detect 2–3 new PULXs, thus doubling the current sample.**

2. Description of the proposed programme

A) Scientific Rationale:

Ultraluminous X-ray sources are off-nuclear objects detected in nearby galaxies with X-ray luminosities in excess of 10^{39} erg s^{-1}, which is the Eddington luminosity ($L_{\rm Edd}$) for a black hole (BH) of 10 $M_\odot$ (Kaaret & Roberts 2017, ARA&A, 55, 303). $L_{\rm Edd}$ sets the upper limit to the accretion luminosity ($L_{\rm acc}$) that a compact object can steadily produce, since for $L_{\rm acc} > L_{\rm Edd}$, the accretion flow is halted by the radiation pressure. For spherical accretion of fully ionized hydrogen, the limit can be written as $L_{\rm Edd} = 4\pi c G M m_{\rm p}/\sigma_{\rm T} \simeq 1.3\times10^{38}(M/M_\odot)$ erg s^{-1}, where $\sigma_{\rm T}$ is the Thomson scattering cross section, $m_{\rm p}$ is the proton mass, and $M/M_\odot$ is the compact object mass in solar masses; for a 1.4 $M_\odot$ neutron star (NS), the maximum accreting luminosity is $\sim 2\times10^{38}$ erg s^{-1}. Since their discovery in the '70s with the Einstein mission (Fabbiano, et al. 1992, ApJS, 80, 531), the high luminosity of ULXs has thus been interpreted as accretion at or above the Eddington luminosity onto BHs of stellar origin (<80–100 $M_\odot$), or onto intermediate-mass (10^3–10^5 $M_\odot$) BHs (Poutanen et al. 2007, MNRAS, 377, 1187; Zampieri et al. 2009, MNRAS 400, 677).

The recent discovery of coherent pulsations with periods between 0.4 s and 1.4 s interval in the X-ray light curves of three distinct ULXs with luminosities in the $10^{40} \div 10^{41}$ erg s^{-1} range, unambiguously associate these ULXs with accreting NSs, therefore a compact object with mass of only $\sim$1.4$M_\odot$ or slightly larger (Bachetti et al 2014, Nature, 514, 202; Fuerst et al 2016, ApJ, 831, L14; Israel et al. 2017a, Science, 355, 817; Israel et al. 2017b, MNRAS, 466, L48; see Figure 1). More importantly, **these X-ray pulsars demonstrates that accreting NSs can achieve extreme luminosities, above 500 times the $L_{\rm Edd}$, which is simply not conceivable in the current accretion models**. A significantly super-Eddington luminosity can be achieved if the magnetic field of the NS is very high due to the reduction of the scattering cross sections: a luminosity of $\sim$500 $L_{\rm Edd}$ would require a field strength of > 10^{15} G (Mushtukov et al 2015, MNRAS, 454, 2539; Dall'Osso 2015, MNRAS 449, 2144). However, the $\sim$1 s rotation of the NS and its magnetosphere would drag matter at the magnetospheric boundary so fast that the centrifugal force would exceed the gravitational force, inhibiting the accretion on the NS surface by the so-called propeller mechanism (Illarionov et al. 1975 A&A 39, 185; Stella et al. 1986, ApJ, 308, 669). This problem can be partially mitigated assuming that the emission is beamed (as expected in pulsars) though, in the most extreme case, an unrealistic beaming factor of 1/100 would be needed. Several new possible scenarios have been proposed to account for the PULX properties and the presence of a strong multipolar magnetic field ($\sim10^{14}$ G) close to the surface of the NSs appears a reasonable way out of the problem (Israel et al. 2017a; Chashkina et al. 2017 ApJ, 308, 669), though "standard" magnetic fields of $\sim10^{12}$ G are not excluded by models (King et al. 2017, MNRAS, 468, L59).

Another important consequence of the discovery of the PULXs is that the nature of many ULXs which have been classified in good faith as accreting black holes due to their high luminosity is now in doubt. An unknown but possibly large fraction of ULXs might host an accreting NS rather than a BH. The answer to this issue will have a strong impact on many topics beyond compact object studies and accretion models. For instance, the existence of IMBHs is a channel for the formation of 10^{4-5} $M_\odot$ BHs, which are thought

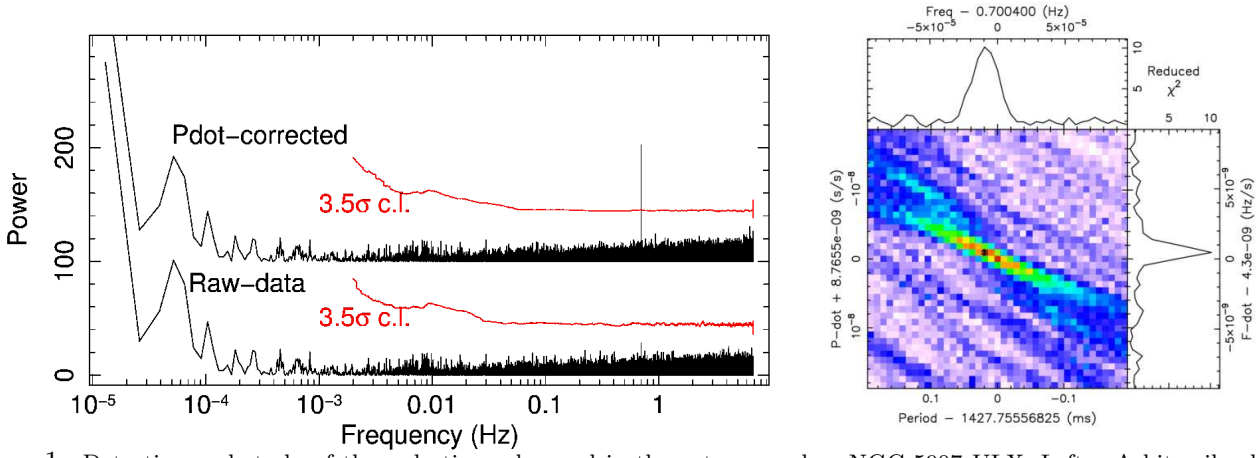

Figure 1: Detection and study of the pulsations observed in the extreme pulsar NGC 5907 ULX. Left – Arbitrarily shifted (along the y-axis) power spectral density (PSD) of the 0.2-12 keV (XMM-Newton) NGC 5907 ULX light curves of the XMM-Newton archival observations carried out on 2003 February 20 before (lower PSD) and after (upper PSD) correcting the photon arrival times for the effect of a strong $\dot{P}$ component ($\sim 9\times10^{-9}$ s s^{-1}). Red curves mark the 3.5σ detection threshold for each PSD. Right – P versus $\dot{P}$ diagram used to refine the timing parameters of the pulsars. The figure clearly shows the importance of the correction: a small variation of $\sim$5% with respect to the inferred $\dot{P}$ value is able to wash out the spin signal.

to be trelevant for the presence of supermassive BHs in quasars at $z > 6$–7 (see Pacucci et al. 2017, ApJ, 835, L36).

Assessing the nature of the compact objects hosted in ULXs is of paramount importance and the discovery of three PULXs demonstrate that we can tackle the problem. In general, the unambiguous identification of the nature of a NS in X-rays is achieved with: **[1]** the detection of rapid coherent signals reflecting the spin period of the NS ("BHs have no hairs" and no X-ray modulation related to its spin is detectable); **[2]** the detection of CRSFs (though less frequently seen) in their energy spectra . However, given the pulsed fractions (in the following we adopt the definition of semi-amplitude of the sinusoid divided by the source average count rate) observed in the PULXs and the intrinsically low intensity of CRSFs (we adopt the equivalent width, EW, measure to define their strength), both approaches need large count-statistics, order of or higher than $\sim$10,000 counts (see below). This is currently available only for about $\sim$13 ULXs (among which are the three PULXs) out of the sample of about 300 known (Earnshaw et al. 2017).

The detection of pulsations (as well as of CRSFs) does not depend on the brightness of the target as much as on the total number of source counts present in the data. In fact, we can detect at 3.5σ confidence level a pulsation in a fainter source by observing it for longer. The 3 PULXs known so far have all PFs above 10%. We show in Figure 2 that, **in order to detect a 10% PF pulsation, at least 10.000 counts are needed**. Counts collected in multiple, far apart observations, will not do the job, because of the orbital motion and the moderate-to-large $\dot{P}$ these sources have ($\dot{P} \sim 10^{-8} \div 10^{-11}$ s s^{-1}), which eventually washes out the signal too much to be corrected with reasonable searches. As of today, there are only 10 additional ULXs in the XMM archive for which such pulsations could have been detected, i.e. with more than 10,000 source counts in a single observation (see Figure 2, left panel, objects above the blue stepped line in the right histogram plot). For other $\sim$10 ULXs, the number of counts (between 2,000 and 8,000 counts) would have allowed the detection of pulsations only for PFs larger than 20%-30%. However, **for the large majority of the ULXs observed with XMM, the number of counts collected is by far too low to allow the detection of such pulsations**.

[1] The process of detecting a signal with respect to the statistical/instrumental noise(s) is carried out by using several different algorithms. Among others are the Fast Fourier Transform (Leahy et al. 1983 ApJ, 266, 160L), which we modified in order to take into account the possible presence of non-Possonian noise components in the power spectra (PSD; Israel & Stella 1996, ApJ, 468, 369), and the Z^2_N periodogram (Buccheri 1988, Space Science Reviews, 49, 197). These two methods are routinely applied since both methods have pro and cons and are somewhat complementary: FFTs are fast but less sensitive to non-sinusoidal signals and/or to time series with few counts (below 100÷200; see Figure 2), while Z^2_N periodograms are more sensitive to faint and/or non-sinusoidal signals but quite CPU-consuming. Fur-

thermore, we modified our searching strategies following the extreme timing properties of two out of the three new pulsators we discovered which possess $\dot{P}$ in the $10^{-(8\div9)}\,\mathrm{s\,s^{-1}}$ and orbital periods in the range of few days. Figure 1 clearly summarizes this issue: without a proper correction the signal of NGC 5907 ULX cannot be detected. Therefore, we developed new pipelines which systematically and automatically sample a multi dimensional grid of timing parameters (up to six: the spin period P, its first derivative $\dot{P}$, the orbital period $P_{\rm orb}$, the semi major axis projection $a_{\rm X}\sin i$, and the time of the ascendent node $T_{\rm node}$). Depending on the number of free parameters and grid step, up to 10^3–10^6 PSDs or periodograms are needed to complete the search analysis for each light curve. In order to speed up this otherwise time-consuming approach, we are routinely relying upon High Performance Computers (HPC; both with guaranteed access or available on a competition basis) ensuring a sensitive signal search over a large sample of light curves, maximizing the parameter resolution and with a relatively short execution time. One of the latest HPC run we are carrying out is dedicated to the search (correcting only for the $\dot{P}$ component) of ULXs signals in the XMM-Newton archive and **no new pulsar has been detected so far while in almost all cases the 3σ upper limits to the PF are above the 10-15% observed in PULXs** (Rodriguez et al. 2017, in prep.). In FFTs, the relation which links the number of counts to the minimum detectable signal PF is given by $PF = \left\{\left[\frac{P_j}{2M}\right]\frac{4}{0.773N_\gamma}\frac{(\pi j/N)^2}{sin^2(\pi j/N)}\right\}^{\frac{1}{2}}$, where P_j is the power in the jth Fourier frequency, N_γ and N the number of counts and bins in the time series, and M the number of averaged FFTs in the final PSD (Leahy et al. 1983). This relation has been applied to all the sources listed in the ULX catalog and plotted in the first panel of Figure 2 (see caption for details): it is evident that, currently, **there are only 10 ULXs** (red dots above the blue stepped line) **with a statistics, within a single pointing, comparable or higher to that of the three PULXs** (magenta dots). **This number (10+3) is only ∼6% of the total known ULX population. More importantly, the incidence of NS among this 6% of ULXs with good statistics is ∼23%; this fraction is the subject of the request**.

[2] The CRSF detection approach, even though less promising in terms of detections (∼40 known CRSFs, out of ∼250 X-ray pulsars), is feasible due to the very high magnetic fields (>10^{14}G) which are

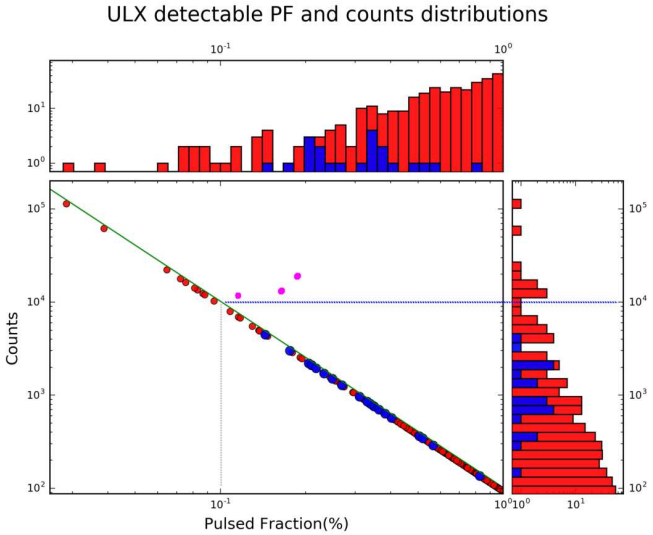

Figure 2: The theoretical pulsed fraction (PF) 3.5σ upper limits for FFTs (green line) versus the maximum recorded counts in a single XMM observation for all the ULXs and super-Eddington (L_X >2×10^{38} erg s^{-1}) off-nuclear sources recorded in the catalog (red filled circle; Earnshaw et al. 2017) and for the sources we are proposing to observe (blue filled circles), where the stepped blue line marks the 10. 000 counts threshold, while the gray line marks the tightest 3.5σ upper limit on PF (∼ 10%) we can infer for the assumed count threshold. The magenta circles show the position of the pulsars in NGC 5907 and NGC 7793 (M82 X-2 pulsations have not been detected with XMM; NuSTAR PFs are in the 10%-20% range). PFs and counts histograms are also plotted above and on the right.

supposed to be responsible for the super-Eddington luminosities together with the high $\dot{P}$ of the pulsars. In fact, the cyclotron energy (in keV) for a particle of mass m in a magnetic field B (in G) is given by $E_B \simeq 11.6/(1+z)(m_e/m)(B/10^{12})$, where $(1+z)^{-1}=(1-2GM_{\rm NS}/(Rc^2))^{1/2}$ ∼0.8 (assuming a neutron star mass of $M_{\rm NS} = 1.4\,M_\odot$ and radius $R_{\rm NS} = 10\,$km) is the gravitational redshift. Proton cyclotron lines fall in the 1–10 keV range if the local magnetic field is in the $10^{14}\div10^{15}$G range, as proposed by some models for the PULXs. Nonetheless, the proton CRSFs are intrinsically less deep and wide than those of electrons (EW in the 0.02÷0.2 keV range), making their detection a difficult task (convincing proof of their detection have been obtained for extremely magnetic isolated NSs at luminosities in the 10^{34}-10^{40} erg s^{-1}range; Ibrahim et al. 2002, ApJ, 574, L51; Tiengo et al. 2013 Nature, 500, 312; Rodriguez 2016, MNRAS, 456, 4145; Borghese 2016 MNRAS, 468, 2975; Borghese et al 2017, MNRAS 468, 2975). Recently, Brightman (et al. 2017, submitted) found a feature, similar to those observed in transient magnetars, in a 12. 000 counts Chandra ACIS spectrum of a L_X ∼10^{40} erg s^{-1}ULX that, if interpreted in terms of proton-CRSF implies a mag-

netic field of B$\sim 10^{15}$, confirming that the proposed strategy is valid. The inspection of a subsample of spectra of 8 ULXs with counts below our threshold (source photons in the 1,000÷6,000 range) confirms that we are not sensitive to any spectral feature with parameters similar to CRSFs detected in isolated NSs (EWs below ∼0.2 keV).

It is evident, from the above considerations, that a major step forward in the study and understanding of ULXs goes through the expansion of the sample of ULXs where a sensitive search for pulsations and CRSFs can be carried out. With this aim we considered the latest updated catalog of ULXs and super-Eddington off-nuclear X-ray sources detected by XMM-Newton in nearby galaxies (Earnshaw et al. 2017; Figure 2 left panel) and filtered objects according to the following criterion: **i)** to collect at least 10,000 photons within one XMM orbit (∼135 ks), and **ii)** maximum visibility time per orbit larger than the requested time to collect the above counts. The final sample includes 20 sources in 10 pointings for a total of 38 sources with super-Eddington luminosities within the EPICs fields of view (FOVs). Condition **i)** is necessary in order to maximize the Nyquist frequency ($0.5\Delta t^{-1}$ where Δt is the time series sampling time) for the signal search while keeping the maximum Fourier resolution (T^{-1}, where T is the observational timespan): the optimal approach for the search of coherent signals. Within the final selected sample we also have sources with $L_{\rm Edd} < L_{\rm X} < 10^{39}$ erg s^{-1}, therefore formally not classified as ULXs. However, under the assumption that a good fraction of ULX are indeed accreting NS, pulsations might be detected also among these sources (though at PFs > 20%); the study of the pulsars properties as a function of luminosity (with the advantage of a good knowledge of the distance contrary to what happens to many Galactic X-ray pulsars) is important for the understanding of accretion processes at super-Eddington regimes.

B) Immediate Objective:

This proposal aims at unveiling the NS population hosted by ULXs or, at least, constraining it. We ask to observe 20 ULXs for long enough to collect at least 10. 000 counts each in a single pointing, thus making it possible to detect such elusive signals/features. For each of the selected targets, the requested exposure will imply an increase in the number of counts in a single observation by at least a factor of two (up to in some cases a factor of 500; see Figure 2). In particular we would try to: **[1]** triple the number of ULXs over which is possible

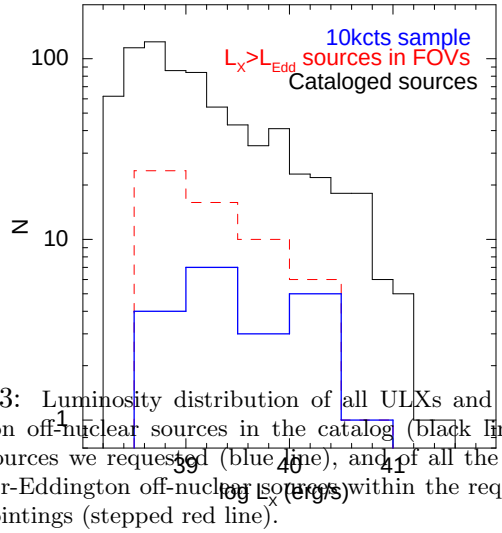

Figure 3: Luminosity distribution of all ULXs and super-Eddington off-nuclear sources in the catalog (black line), of the 20 sources we requested (blue line), and of all the ULXs and super-Eddington off-nuclear sources within the requested XMM pointings (stepped red line).

to carry out sensitive searches (PFs > 10%) for the NS spin period (with characteristics similar to those of known PULXs): in principle ∼4÷6 new pulsars might be discovered within the project, though we may also expect a ∼20-25% of AGN contamination in our ULX sample (Earnshaw et al. 2017) which reduces the above numbers to a more realistic ∼2÷3 PULXs, therefore almost dubling the current known population; **[2]** searching, within the new sample of 20 objects, for proton CRSFs (with EWs > 0.15 keV) expected in the 1–10 keV range for magnetic fields (close to the surface) of 10^{14}÷10^{15}G, as recently observed in one ULX; **[3]** searching for new signals with PFs >20% in a sample of 38 ULXs and off-nuclear super-Eddington objects. Additionally, we expect **[4]** to use the results as input for the accretion models and **[5]** to carry out population studies, for the first time with a good statistics for about 20 sources, based on their timing (such as variability) and spectral properties (hard versus soft; Pintore et al. 2017 ApJ, 836, 113). These will be used as reliable template to the simulations/theoretical models aimed at accounting for the formation and evolution of ULXs (Wiktorowicz et al. 2017, ApJ, 846, 17).

3. Observing time, feasibility and visibility

All the fields we are requesting have been already observed by XMM at least one time, therefore the observations are feasible. Additionally, the main target of each pointing has been selected based on the request **i)** of collecting at least 10,000 counts in one XMM orbit based on the average count rates of previous pointings (note that to infer count rates we assumed

the counts and effective exposures as reported in the 3XMM catalog which may differ from the nominal observational time). All the selected sources are resolved and sufficently far away from nearby objects. We note that only 4 targets show a moderate level of variability among archival pointings (within a factor of about 3); in these cases we adopt the lowest available count rate in order to infer the needed exposure to collect 10,000 counts. We emphasize that there is no special motivation to include or exclude *a priori* persistent or variable objects from the selected targets; all three PULXs show off states which are however fairly rare, while pulsations in NGC 7793 P13 has been always detected (but one), in NGC 5907 pulsations have been detected ∼50% of the times and in M82 X-2, which shows a bimodal luminosity distribution, pulsations have been detected only once. We emphasize that even if one or two sources would not be detected or found in a low state preventing the main science goal, also it would be itself a very interesting result allowing to tag them as likely X-ray pulsars. We also checked, by means of simulations carried out with XSPEC and PIMMS, that our EW sensitivity (3σ) to CRSFs (input parameters taken from the proton-CRSFs detected so far) for a 10. 000 count spectrum is >0.15-0.25 keV depending on the energy and depth of the feature, and on the shape of the spectral continuum (Pintore et al. 2017).

We note that moderately intense proton flares (in 25% ot the datasets) will not strongly affect our search sensitivity for periods shorter than ∼10s, while can effect (lower statistics) the above reported sensitivities for CRSFs. In all cases we will adopt the EPICs as follows: pn in full frame (73ms time resolution, corresponding to a minimum period of ∼150ms for the signal search), and MOSs in full frame (2.6s time resolution). Roll angles similar to those of the archival pointings will be requested. The targets have been checked in the visibility tool (criteria **ii)** of our selection). We also checked that the requested targets are not present in the latest approved Chandra (AO17)and XMM (OA16) lists. Table 1 summarizes our targets ordered according to our priorities (defined by taking into account several parameters: number of ULXs and super-Eddington sources in the FOV, luminosity, etc.). **The total request is 972 ks for 10 pointings**.

Finally, we have institutional access at up to 12 million core-hours per year (Dalma, Uni. NY, Dubai) to perform the blind accelerated searches for signals, together with the parallelized software infrastructure for accelerated searches which has been directly developed by members of the proposal within the EXTraS project (www.extras-fp7.eu).

Table 1: Priority ordered Targets

Target	T_{exp} (ks)	$L_X \cdot 10^{39}$ (erg s^{-1})	d (Mpc)
J121839.3+47	130	2.4	3.6
J121847.6+47	”	1.5	6.0
J121856.3+47	”	0.5	”
J121857.8+47	”	0.7	”
J123030.5+41	114	4.7	8.8
J123030.8+414	”	4.4	”
J123043.1+41	”	1.1	5.3
J141239.4-65	130	16.3	6.8
J141253.5-65	”	1.3	”
J230205.7-39	79	13.8	17.4
J230209.9-39	”	14.0	”
J140314.2+54	100	3.1	7.0
J140414.0+54	”	1.1	”
J230457.6+12	68	53.8	28.2
J231856.2-42	106	7.1	14.6
J122809.3+44	84	0.4	4.0
J122810.9+44	”	0.5	”
J122817.8+44	”	1.2	”
J133719.8-29	71	2.7	4.5
J102842.9+68	87	0.6	3.6
Total	969		

4. Report on the last use of XMM-Newton

ToO observations of SGR 1935+2154 (2014 and 2015) have been published (Israel et al. 2016, MNRAS, 460, 2008). Observation of AX J1910.7+0917 (2016) have been published (Sidoli et al. 2017, 469, 3056). We are still working on CATS J193437.8 (in 2013), CATS J174728.1, CATS J225355.1 and CATS J112347.4 (in 2014). Data from the observational campaign of ULXs (2013) have been published (e.g. Bachetti et al. 2013, ApJ, 778, 163). Data from the LP (AO16) on NGC 5907 ULX are still being obtained and analyses are in progress.

5. Most relevant proposer's publications

Bachetti M., Harrison F., et al. Nature, 2014, 414, 202: *An ultraluminous X-ray source powered by an accreting neutron star*

Israel G.L., Belfiore A., et al. Science, 2017, 355, 817: *An accreting pulsar with extreme properties drives an ultraluminous X-ray source in NGC 5907*

Füerst F., Walton D., et al. 2016, ApJ, 831, L14:*Discovery of Coherent Pulsations from the Ultraluminous X-Ray Source NGC 7793 P13*

Israel G.L., Papitto A., et al. MNRAS, 2017, 466, L48:*Discovery of a 0.42-s pulsar in the ultraluminous X-ray source NGC 7793 P13*